

```

from pyspark.sql import SparkSession
from pyspark.sql.functions import *
from pyspark.sql.window import Window
spark =
SparkSession.builder.appName("PySparkAssignment").getOrCreate()

orders_data = [
    (101, 1, "Laptop", "Electronics", 50000, 1, "2025-01-10",
"Hyderabad"),
    (102, 2, "Mobile", "Electronics", 30000, 2, "2025-01-11",
"Bangalore"),
    (103, 3, "TV", "Electronics", 45000, 1, "2025-01-12", "Chennai"),
    (104, 1, "Shoes", "Fashion", 3000, 2, "2025-01-13", "Hyderabad"),
    (105, 4, "Watch", "Fashion", 5000, 1, "2025-01-14", "Mumbai"),
    (106, 5, "Refrigerator", "Appliances", 35000, 1, "2025-01-15",
"Delhi"),
    (107, 2, "AC", "Appliances", 40000, 1, "2025-01-16", "Bangalore"),
    (108, 6, "Mobile", "Electronics", 25000, 3, "2025-01-17", "Pune"),
    (109, 3, "Shoes", "Fashion", 3500, 2, "2025-01-18", "Chennai"),
    (110, 7, "Laptop", "Electronics", 60000, 1, "2025-01-19",
"Kolkata"),
    (111, 8, "Watch", "Fashion", 4500, 2, "2025-01-20", "Delhi"),
    (112, 4, "Mobile", "Electronics", 28000, 1, "2025-01-21",
"Mumbai"),
    (113, 9, "TV", "Electronics", 55000, 1, "2025-01-22",
"Hyderabad"),
    (114, 10, "AC", "Appliances", 42000, 1, "2025-01-23", "Pune"),
    (115, 5, "Shoes", "Fashion", 2500, 4, "2025-01-24", "Delhi")
]

orders_cols = ["order_id", "customer_id", "product",
               "category", "price", "quantity",
               "order_date", "city"]

orders_df = spark.createDataFrame(orders_data, orders_cols)
orders_df.show()

```

```

+-----+-----+-----+-----+-----+-----+
+-----+-----+
|order_id|customer_id|    product|    category|price|quantity|
order_date|    city|
+-----+-----+-----+-----+-----+-----+
+-----+-----+
|    101|          1|    Laptop|Electronics|50000|          1|2025-01-
10|Hyderabad|
|    102|          2|    Mobile|Electronics|30000|          2|2025-01-
11|Bangalore|
|    103|          3|         TV|Electronics|45000|          1|2025-01-
12|  Chennai|

```

13	Hyderabad	104	1	Shoes	Fashion	3000	2	2025-01-
14	Mumbai	105	4	Watch	Fashion	5000	1	2025-01-
15	Delhi	106	5	Refrigerator	Appliances	35000	1	2025-01-
16	Bangalore	107	2	AC	Appliances	40000	1	2025-01-
17	Pune	108	6	Mobile	Electronics	25000	3	2025-01-
18	Chennai	109	3	Shoes	Fashion	3500	2	2025-01-
19	Kolkata	110	7	Laptop	Electronics	60000	1	2025-01-
20	Delhi	111	8	Watch	Fashion	4500	2	2025-01-
21	Mumbai	112	4	Mobile	Electronics	28000	1	2025-01-
22	Hyderabad	113	9	TV	Electronics	55000	1	2025-01-
23	Pune	114	10	AC	Appliances	42000	1	2025-01-
24	Delhi	115	5	Shoes	Fashion	2500	4	2025-01-

+-----+-----+-----+-----+-----+-----+
+-----+-----+

```
orders_df.printSchema()
```

```
root
|-- order_id: long (nullable = true)
|-- customer_id: long (nullable = true)
|-- product: string (nullable = true)
|-- category: string (nullable = true)
|-- price: long (nullable = true)
|-- quantity: long (nullable = true)
|-- order_date: string (nullable = true)
|-- city: string (nullable = true)
```

```
orders_df.count()
```

```
15
```

Section-1:Basic Transformations

```
orders_df.select("order_id","product","price").show()
```

order_id	product	price
101	Laptop	50000
102	Mobile	30000
103	TV	45000
104	Shoes	3000
105	Watch	5000
106	Refrigerator	35000
107	AC	40000
108	Mobile	25000
109	Shoes	3500
110	Laptop	60000
111	Watch	4500
112	Mobile	28000
113	TV	55000
114	AC	42000
115	Shoes	2500

```
q2_df = orders_df.withColumnRenamed(
    "product",
    "product_name"
)
```

```
q2_df.printSchema()
```

```
root
|-- order_id: long (nullable = true)
|-- customer_id: long (nullable = true)
|-- product_name: string (nullable = true)
|-- category: string (nullable = true)
|-- price: long (nullable = true)
|-- quantity: long (nullable = true)
|-- order_date: string (nullable = true)
|-- city: string (nullable = true)
```

```
orders_df.printSchema()
```

```
root
|-- order_id: long (nullable = true)
|-- customer_id: long (nullable = true)
|-- product: string (nullable = true)
|-- category: string (nullable = true)
|-- price: long (nullable = true)
|-- quantity: long (nullable = true)
|-- order_date: string (nullable = true)
|-- city: string (nullable = true)
```

```
q3_df = orders_df.withColumn(
    "total_amount",
    col("price") * col("quantity")
)
```

```
q3_df.show()
```

```
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+
|order_id|customer_id|    product|    category|price|quantity|
order_date|    city|total_amount|
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+
|    101|          1|    Laptop|Electronics|50000|    1|2025-01-
10|Hyderabad|    50000|
|    102|          2|    Mobile|Electronics|30000|    2|2025-01-
11|Bangalore|    60000|
|    103|          3|         TV|Electronics|45000|    1|2025-01-
12|  Chennai|    45000|
|    104|          1|    Shoes|    Fashion| 3000|    2|2025-01-
13|Hyderabad|    6000|
|    105|          4|    Watch|    Fashion| 5000|    1|2025-01-
14|  Mumbai|    5000|
|    106|          5|Refrigerator| Appliances|35000|    1|2025-01-
15|   Delhi|    35000|
|    107|          2|         AC| Appliances|40000|    1|2025-01-
16|Bangalore|    40000|
|    108|          6|    Mobile|Electronics|25000|    3|2025-01-
17|   Pune|    75000|
|    109|          3|    Shoes|    Fashion| 3500|    2|2025-01-
18|  Chennai|    7000|
|    110|          7|    Laptop|Electronics|60000|    1|2025-01-
19| Kolkata|    60000|
|    111|          8|    Watch|    Fashion| 4500|    2|2025-01-
20|   Delhi|    9000|
|    112|          4|    Mobile|Electronics|28000|    1|2025-01-
21|  Mumbai|    28000|
|    113|          9|         TV|Electronics|55000|    1|2025-01-
22|Hyderabad|    55000|
|    114|         10|         AC| Appliances|42000|    1|2025-01-
23|   Pune|    42000|
|    115|          5|    Shoes|    Fashion| 2500|    4|2025-01-
24|   Delhi|   10000|
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+
```

```
q4_df=q3_df.withColumn(
    "gstamount",
```

```
col("total_amount")*0.18)
q4_df.show()
```

```
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+
|order_id|customer_id|    product|    category|price|quantity|
order_date|    city|total_amount|gstamount|
+-----+-----+-----+-----+
+-----+-----+-----+-----+
|    101|          1|    Laptop|Electronics|50000|    1|2025-01-
10|Hyderabad|    50000|    9000.0|
|    102|          2|    Mobile|Electronics|30000|    2|2025-01-
11|Bangalore|    60000|    10800.0|
|    103|          3|        TV|Electronics|45000|    1|2025-01-
12|Chennai|    45000|    8100.0|
|    104|          1|    Shoes|    Fashion| 3000|    2|2025-01-
13|Hyderabad|    6000|    1080.0|
|    105|          4|    Watch|    Fashion| 5000|    1|2025-01-
14|Mumbai|    5000|    900.0|
|    106|          5|Refrigerator|Appliances|35000|    1|2025-01-
15|Delhi|    35000|    6300.0|
|    107|          2|        AC|Appliances|40000|    1|2025-01-
16|Bangalore|    40000|    7200.0|
|    108|          6|    Mobile|Electronics|25000|    3|2025-01-
17|Pune|    75000|    13500.0|
|    109|          3|    Shoes|    Fashion| 3500|    2|2025-01-
18|Chennai|    7000|    1260.0|
|    110|          7|    Laptop|Electronics|60000|    1|2025-01-
19|Kolkata|    60000|    10800.0|
|    111|          8|    Watch|    Fashion| 4500|    2|2025-01-
20|Delhi|    9000|    1620.0|
|    112|          4|    Mobile|Electronics|28000|    1|2025-01-
21|Mumbai|    28000|    5040.0|
|    113|          9|        TV|Electronics|55000|    1|2025-01-
22|Hyderabad|    55000|    9900.0|
|    114|         10|        AC|Appliances|42000|    1|2025-01-
23|Pune|    42000|    7560.0|
|    115|          5|    Shoes|    Fashion| 2500|    4|2025-01-
24|Delhi|    10000|    1800.0|
+-----+-----+-----+-----+
+-----+-----+-----+-----+
```

```
q4_df = orders_df.withColumn(
    "discount",
    when(col("category") == "Electronics", 10)
    .when(col("category") == "Fashion", 5)
    .when(col("category") == "Appliances", 8)
    .otherwise(0)
).show()
```

```

+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+
|order_id|customer_id|    product|    category|price|quantity|
order_date|    city|discount|
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+
|    101|    1|    Laptop|Electronics|50000|    1|2025-01-
10|Hyderabad|    10|
|    102|    2|    Mobile|Electronics|30000|    2|2025-01-
11|Bangalore|    10|
|    103|    3|    TV|Electronics|45000|    1|2025-01-
12|Chennai|    10|
|    104|    1|    Shoes|    Fashion| 3000|    2|2025-01-
13|Hyderabad|    5|
|    105|    4|    Watch|    Fashion| 5000|    1|2025-01-
14|Mumbai|    5|
|    106|    5|Refrigerator|Appliances|35000|    1|2025-01-
15|Delhi|    8|
|    107|    2|    AC|Appliances|40000|    1|2025-01-
16|Bangalore|    8|
|    108|    6|    Mobile|Electronics|25000|    3|2025-01-
17|Pune|    10|
|    109|    3|    Shoes|    Fashion| 3500|    2|2025-01-
18|Chennai|    5|
|    110|    7|    Laptop|Electronics|60000|    1|2025-01-
19|Kolkata|    10|
|    111|    8|    Watch|    Fashion| 4500|    2|2025-01-
20|Delhi|    5|
|    112|    4|    Mobile|Electronics|28000|    1|2025-01-
21|Mumbai|    10|
|    113|    9|    TV|Electronics|55000|    1|2025-01-
22|Hyderabad|    10|
|    114|    10|    AC|Appliances|42000|    1|2025-01-
23|Pune|    8|
|    115|    5|    Shoes|    Fashion| 2500|    4|2025-01-
24|Delhi|    5|
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+

```

```

q5_df = orders_df.withColumn(
    "order_date",
    to_date(col("order_date"))
)

```

```

q5_df.show()

```

```

+-----+-----+-----+-----+-----+-----+
+-----+-----+
|order_id|customer_id|    product|    category|price|quantity|

```

order_id	order_date	city	product	category	price	quantity	order_date
10	2025-01-10	Hyderabad	Laptop	Electronics	50000	1	2025-01-10
11	2025-01-11	Bangalore	Mobile	Electronics	30000	2	2025-01-11
12	2025-01-12	Chennai	TV	Electronics	45000	1	2025-01-12
13	2025-01-13	Hyderabad	Shoes	Fashion	3000	2	2025-01-13
14	2025-01-14	Mumbai	Watch	Fashion	5000	1	2025-01-14
15	2025-01-15	Delhi	Refrigerator	Appliances	35000	1	2025-01-15
16	2025-01-16	Bangalore	AC	Appliances	40000	1	2025-01-16
17	2025-01-17	Pune	Mobile	Electronics	25000	3	2025-01-17
18	2025-01-18	Chennai	Shoes	Fashion	3500	2	2025-01-18
19	2025-01-19	Kolkata	Laptop	Electronics	60000	1	2025-01-19
20	2025-01-20	Delhi	Watch	Fashion	4500	2	2025-01-20
21	2025-01-21	Mumbai	Mobile	Electronics	28000	1	2025-01-21
22	2025-01-22	Hyderabad	TV	Electronics	55000	1	2025-01-22
23	2025-01-23	Pune	AC	Appliances	42000	1	2025-01-23
24	2025-01-24	Delhi	Shoes	Fashion	2500	4	2025-01-24

```
q5_df.printSchema()
```

```
root
|-- order_id: long (nullable = true)
|-- customer_id: long (nullable = true)
|-- product: string (nullable = true)
|-- category: string (nullable = true)
|-- price: long (nullable = true)
|-- quantity: long (nullable = true)
|-- order_date: date (nullable = true)
|-- city: string (nullable = true)
```

```
q5_df.select(
    year("order_date")
).show()
```

```
+-----+
|year(order_date)|
+-----+
|                |
|                |
|                |
|                |
|                |
|                |
|                |
|                |
|                |
|                |
|                |
|                |
|                |
|                |
|                |
|                |
|                |
+-----+
```

```
orders_df.filter(col("city")=="Hyderabad").show()
```

```
+-----+-----+-----+-----+-----+-----+-----+
+-----+
|order_id|customer_id|product|category|price|quantity|order_date|
city|
+-----+-----+-----+-----+-----+-----+-----+
+-----+
|      101|          1| Laptop|Electronics|50000|          1|2025-01-10|
Hyderabad|
|      104|          1|  Shoes|    Fashion| 3000|          2|2025-01-13|
Hyderabad|
|      113|          9|    TV|Electronics|55000|          1|2025-01-22|
Hyderabad|
+-----+-----+-----+-----+-----+-----+-----+
+-----+
```

```
orders_df.filter(
    col("quantity") > 2
).show()
```

```
+-----+-----+-----+-----+-----+-----+-----+
+-----+
|order_id|customer_id|product|category|price|quantity|order_date|
city|
```



```

+-----+-----+-----+-----+-----+-----+
+-----+
|      108|          6| Mobile|Electronics|25000|          3|2025-01-17|
Pune|
|      115|          5|  Shoes|    Fashion| 2500|          4|2025-01-24|
Delhi|
+-----+-----+-----+-----+-----+-----+
+-----+

```

```
orders_df.filter(col("category") == "Electronics").show()
```

```

+-----+-----+-----+-----+-----+-----+
+-----+
|order_id|customer_id|product|  category|price|quantity|order_date|
city|
+-----+-----+-----+-----+-----+-----+
+-----+
|      101|          1| Laptop|Electronics|50000|          1|2025-01-10|
Hyderabad|
|      102|          2| Mobile|Electronics|30000|          2|2025-01-11|
Bangalore|
|      103|          3|    TV|Electronics|45000|          1|2025-01-12|
Chennai|
|      108|          6| Mobile|Electronics|25000|          3|2025-01-17|
Pune|
|      110|          7| Laptop|Electronics|60000|          1|2025-01-19|
Kolkata|
|      112|          4| Mobile|Electronics|28000|          1|2025-01-21|
Mumbai|
|      113|          9|    TV|Electronics|55000|          1|2025-01-22|
Hyderabad|
+-----+-----+-----+-----+-----+-----+
+-----+

```

```

q9_df = orders_df.withColumn(
    "category_upper",
    upper(col("category"))
)

```

```
q9_df.show()
```

```

+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+
|order_id|customer_id|  product|  category|price|quantity|
order_date|      city|category_upper|
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+
|      101|          1|    Laptop|Electronics|50000|          1|2025-01-

```

10	Hyderabad	ELECTRONICS					
	102	2	Mobile	Electronics	30000	2	2025-01-
11	Bangalore	ELECTRONICS					
	103	3	TV	Electronics	45000	1	2025-01-
12	Chennai	ELECTRONICS					
	104	1	Shoes	Fashion	3000	2	2025-01-
13	Hyderabad	FASHION					
	105	4	Watch	Fashion	5000	1	2025-01-
14	Mumbai	FASHION					
	106	5	Refrigerator	Appliances	35000	1	2025-01-
15	Delhi	APPLIANCES					
	107	2	AC	Appliances	40000	1	2025-01-
16	Bangalore	APPLIANCES					
	108	6	Mobile	Electronics	25000	3	2025-01-
17	Pune	ELECTRONICS					
	109	3	Shoes	Fashion	3500	2	2025-01-
18	Chennai	FASHION					
	110	7	Laptop	Electronics	60000	1	2025-01-
19	Kolkata	ELECTRONICS					
	111	8	Watch	Fashion	4500	2	2025-01-
20	Delhi	FASHION					
	112	4	Mobile	Electronics	28000	1	2025-01-
21	Mumbai	ELECTRONICS					
	113	9	TV	Electronics	55000	1	2025-01-
22	Hyderabad	ELECTRONICS					
	114	10	AC	Appliances	42000	1	2025-01-
23	Pune	APPLIANCES					
	115	5	Shoes	Fashion	2500	4	2025-01-
24	Delhi	FASHION					
+-----+-----+-----+-----+-----+-----+-----+-----							
+-----+-----+-----+-----+							

```
q10=orders_df.drop("city").show()
```

+-----+-----+-----+-----+-----+-----+-----+-----							
+-----+							
	order_id	customer_id	product	category	price	quantity	order_date
+-----+-----+-----+-----+-----+-----+-----+-----							
+-----+							
	101	1	Laptop	Electronics	50000	1	2025-01-
10							
	102	2	Mobile	Electronics	30000	2	2025-01-
11							
	103	3	TV	Electronics	45000	1	2025-01-
12							
	104	1	Shoes	Fashion	3000	2	2025-01-
13							
	105	4	Watch	Fashion	5000	1	2025-01-

```

14|
| 106| 5|Refrigerator| Appliances|35000| 1|2025-01-
15|
| 107| 2| AC| Appliances|40000| 1|2025-01-
16|
| 108| 6| Mobile|Electronics|25000| 3|2025-01-
17|
| 109| 3| Shoes| Fashion| 3500| 2|2025-01-
18|
| 110| 7| Laptop|Electronics|60000| 1|2025-01-
19|
| 111| 8| Watch| Fashion| 4500| 2|2025-01-
20|
| 112| 4| Mobile|Electronics|28000| 1|2025-01-
21|
| 113| 9| TV|Electronics|55000| 1|2025-01-
22|
| 114| 10| AC| Appliances|42000| 1|2025-01-
23|
| 115| 5| Shoes| Fashion| 2500| 4|2025-01-
24|
+-----+-----+-----+-----+-----+-----+
+-----+

```

```

customers_data = [
    (1,"Sravan",28,"Hyderabad"),
    (2,"Rahul",30,"Bangalore"),
    (3,"Priya",27,"Chennai"),
    (4,"Sneha",35,"Mumbai"),
    (5,"Arjun",32,"Delhi"),
    (6,"Kiran",29,"Pune"),
    (7,"Amit",40,"Kolkata"),
    (8,"Meena",26,"Delhi"),
    (9,"Ravi",31,"Hyderabad"),
    (10,"Pooja",33,"Pune")
]

customers_cols = ["customer_id","customer_name","age","city"]

customers_df = spark.createDataFrame(customers_data, customers_cols)

customers_df.show()
customers_df.printSchema()
customers_df.describe().show()

+-----+-----+-----+-----+
|customer_id|customer_name|age|city|
+-----+-----+-----+-----+
| 1| Sravan| 28|Hyderabad|

```

2	Rahul	30	Bangalore
3	Priya	27	Chennai
4	Sneha	35	Mumbai
5	Arjun	32	Delhi
6	Kiran	29	Pune
7	Amit	40	Kolkata
8	Meena	26	Delhi
9	Ravi	31	Hyderabad
10	Pooja	33	Pune

```
root
```

```
-- customer_id: long (nullable = true)
-- customer_name: string (nullable = true)
-- age: long (nullable = true)
-- city: string (nullable = true)
```

summary	customer_id	customer_name	age	city
count	10	10	10	10
mean	5.5	NULL	31.1	NULL
stddev	3.0276503540974917	NULL	4.175324338699132	NULL
min	1	Amit	26	Bangalore
max	10	Sravan	40	Pune

```
customers_df.count()
```

```
10
```

Section-2:Filter Transformations (11-15)

```
orders_enriched_df = orders_df.withColumn(
    "total_amount",
    col("price") * col("quantity")
)
```

```
q11_df = orders_enriched_df.filter(
    col("total_amount") > 40000
)
```

```
q11_df.show()
```

order_id	customer_id	product	category	price	quantity	order_date	city	total_amount
----------	-------------	---------	----------	-------	----------	------------	------	--------------

order_id	customer_id	product	category	price	quantity	order_date
101	1	Laptop	Electronics	50000	1	2025-01-10
Hyderabad	50000					
102	2	Mobile	Electronics	30000	2	2025-01-11
Bangalore	60000					
103	3	TV	Electronics	45000	1	2025-01-12
Chennai	45000					
108	6	Mobile	Electronics	25000	3	2025-01-17
Pune	75000					
110	7	Laptop	Electronics	60000	1	2025-01-19
Kolkata	60000					
113	9	TV	Electronics	55000	1	2025-01-22
Hyderabad	55000					
114	10	AC	Appliances	42000	1	2025-01-23
Pune	42000					

```
orders_enriched_df = orders_enriched_df.withColumn(
    "order_date",
    to_date(col("order_date"))
)
```

```
q12_df = orders_enriched_df.filter(
    col("order_date").between(
        "2025-01-15",
        "2025-01-22"
    )
)
```

```
q12_df.show()
```

order_id	customer_id	product	category	price	quantity	order_date
106	5	Refrigerator	Appliances	35000	1	2025-01-15
Delhi	35000					
107	2	AC	Appliances	40000	1	2025-01-16
Bangalore	40000					
108	6	Mobile	Electronics	25000	3	2025-01-17
Pune	75000					
109	3	Shoes	Fashion	3500	2	2025-01-18
Chennai	7000					
110	7	Laptop	Electronics	60000	1	2025-01-19
Kolkata	60000					
111	8	Watch	Fashion	4500	2	2025-01-20

```

20|    Delhi|    9000|
|    112|    4|    Mobile|Electronics|28000|    1|2025-01-
21|    Mumbai|    28000|
|    113|    9|    TV|Electronics|55000|    1|2025-01-
22|Hyderabad|    55000|
+-----+-----+-----+-----+-----+
+-----+-----+

```

```

q13_df = customers_df.filter(
    col("age") > 30
)

```

```

q13_df.show()

```

```

+-----+-----+-----+-----+
|customer_id|customer_name|age|    city|
+-----+-----+-----+-----+
|    4|    Sneha| 35|    Mumbai|
|    5|    Arjun| 32|    Delhi|
|    7|    Amit| 40|    Kolkata|
|    9|    Ravi| 31|Hyderabad|
|   10|    Pooja| 33|    Pune|
+-----+-----+-----+-----+

```

```

q14_df = orders_df.filter(
    col("product").startswith("M")
)

```

```

q14_df.show()

```

```

+-----+-----+-----+-----+-----+-----+-----+
+-----+
|order_id|customer_id|product|    category|price|quantity|order_date|
city|
+-----+-----+-----+-----+-----+-----+-----+
+-----+
|    102|    2| Mobile|Electronics|30000|    2|2025-01-11|
Bangalore|
|    108|    6| Mobile|Electronics|25000|    3|2025-01-17|
Pune|
|    112|    4| Mobile|Electronics|28000|    1|2025-01-21|
Mumbai|
+-----+-----+-----+-----+-----+-----+-----+
+-----+

```

```

q15_df = orders_df.filter(
    col("city").isin(
        "Hyderabad",

```

```

        "Bangalore"
    )
)

q15_df.show()

+-----+-----+-----+-----+-----+-----+-----+
+-----+
|order_id|customer_id|product|category|price|quantity|order_date|city|
+-----+-----+-----+-----+-----+-----+-----+
+-----+
|      101|          1| Laptop|Electronics|50000|          1|2025-01-10|Hyderabad|
|      102|          2| Mobile|Electronics|30000|          2|2025-01-11|Bangalore|
|      104|          1|  Shoes|    Fashion| 3000|          2|2025-01-13|Hyderabad|
|      107|          2|    AC| Appliances|40000|          1|2025-01-16|Bangalore|
|      113|          9|    TV|Electronics|55000|          1|2025-01-22|Hyderabad|
+-----+-----+-----+-----+-----+-----+-----+
+-----+

```

Section 3: Aggregations (16-22)

```

from pyspark.sql.functions import sum

orders_enriched_df.select(
    sum("total_amount").alias("total_sales")
).show()

+-----+
|total_sales|
+-----+
|      527000|
+-----+

orders_enriched_df.groupBy("category") \
    .agg(
        sum("quantity").alias("total_quantity")
    ) \
    .show()

+-----+-----+
|category|total_quantity|
+-----+-----+

```

category	count
Electronics	10
Fashion	11
Appliances	3

```
from pyspark.sql.functions import avg

orders_enriched_df.groupBy("category") \
    .agg(
        avg("price").alias("avg_price")
    ) \
    .show()
```

category	avg_price
Electronics	41857.142857142855
Fashion	3700.0
Appliances	39000.0

```
orders_enriched_df.select(
    max("price")
).show()
orders_enriched_df.orderBy(
    col("price").desc()
).show(1)
```

max(price)
60000

order_id	customer_id	product	category	price	quantity	order_date	city	total_amount
110	7	Laptop	Electronics	60000	1	2025-01-19	Kolkata	60000

only showing top 1 row

```
orders_enriched_df.orderBy(
    col("price").asc()
).show(1)
```



```

+-----+-----+-----+-----+-----+-----+-----+
+-----+
|order_id|customer_id|product|category|price|quantity|order_date|
city|total_amount|
+-----+-----+-----+-----+-----+-----+-----+
+-----+
|      115|          5|  Shoes| Fashion| 2500|          4|2025-01-24|
Delhi|      10000|
+-----+-----+-----+-----+-----+-----+-----+
+-----+
only showing top 1 row

```

```

from pyspark.sql.functions import count

orders_enriched_df.groupBy("city") \
    .agg(
        count("*").alias("total_orders")
    ) \
    .show()

```

```

+-----+-----+
|      city|total_orders|
+-----+-----+
|Hyderabad|          3|
|Bangalore|          2|
|Chennai|          2|
|Mumbai|          2|
|Delhi|          3|
|Pune|          2|
|Kolkata|          1|
+-----+-----+

```

```

orders_enriched_df.groupBy("category") \
    .agg(
        sum("total_amount").alias("revenue")
    ) \
    .orderBy(
        col("revenue").desc()
    ) \
    .show(1)

```

```

+-----+-----+
|category|revenue|
+-----+-----+
|Electronics| 373000|
+-----+-----+
only showing top 1 row

```

Section 4: GroupBy Transformations (23-28)

```
orders_df.groupBy("category") \
    .count() \
    .show()
```

```
+-----+-----+
| category|count|
+-----+-----+
|Electronics|    7|
|   Fashion|    5|
|Appliances|    3|
+-----+-----+
```

```
orders_df.groupBy("city") \
    .agg(
        avg("quantity").alias("avg_quantity")
    ) \
    .show()
```

```
+-----+-----+
|    city| avg_quantity|
+-----+-----+
|Hyderabad|1.3333333333333333|
|Bangalore|          1.5|
|Chennai|          1.5|
|Mumbai|          1.0|
|Delhi|2.3333333333333335|
|Pune|          2.0|
|Kolkata|          1.0|
+-----+-----+
```

```
orders_enriched_df.groupBy("city") \
    .agg(
        sum("total_amount").alias("revenue")
    ) \
    .show()
```

```
+-----+-----+
|    city|revenue|
+-----+-----+
|Hyderabad| 111000|
|Bangalore| 100000|
|Chennai|  52000|
|Mumbai|  33000|
|Delhi|  54000|
|Pune| 117000|
|Kolkata|  60000|
+-----+-----+
```

```
orders_enriched_df.groupBy("customer_id") \
  .agg(
    sum("total_amount").alias("customer_revenue")
  ) \
  .show()
```

customer_id	customer_revenue
1	56000
2	100000
3	52000
4	33000
5	45000
6	75000
7	60000
8	9000
9	55000
10	42000

```
orders_enriched_df.groupBy("customer_id") \
  .agg(
    sum("total_amount").alias("customer_revenue")
  ) \
  .orderBy(
    col("customer_revenue").desc()
  ) \
  .limit(3) \
  .show()
```

customer_id	customer_revenue
2	100000
6	75000
7	60000

```
orders_enriched_df.groupBy("city") \
  .agg(
    sum("total_amount").alias("sales")
  ) \
  .orderBy(
    col("sales").desc()
  ) \
  .show(1)
```

```
+-----+-----+
|city| sales|
+-----+-----+
|Pune|117000|
+-----+-----+
only showing top 1 row
```

Section 5: Join Transformations (29-34)

```
joined_df = orders_enriched_df.join(
    customers_df,
    on="customer_id",
    how="inner"
)
```

```
joined_df.show()
```

```
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
|customer_id|order_id|    product|    category|price|quantity|
order_date|    city|total_amount|customer_name|age|    city|
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
|          1|    101|    Laptop|Electronics|50000|          1|2025-01-
10|Hyderabad|    50000|    Sravan| 28|Hyderabad|
|          2|    102|    Mobile|Electronics|30000|          2|2025-01-
11|Bangalore|    60000|    Rahul| 30|Bangalore|
|          3|    103|    TV|Electronics|45000|          1|2025-01-
12|Chennai|    45000|    Priya| 27|Chennai|
|          1|    104|    Shoes|    Fashion| 3000|          2|2025-01-
13|Hyderabad|    6000|    Sravan| 28|Hyderabad|
|          4|    105|    Watch|    Fashion| 5000|          1|2025-01-
14|Mumbai|    5000|    Sneha| 35|Mumbai|
|          5|    106|Refrigerator|Appliances|35000|          1|2025-01-
15|Delhi|    35000|    Arjun| 32|Delhi|
|          2|    107|    AC|Appliances|40000|          1|2025-01-
16|Bangalore|    40000|    Rahul| 30|Bangalore|
|          6|    108|    Mobile|Electronics|25000|          3|2025-01-
17|Pune|    75000|    Kiran| 29|Pune|
|          3|    109|    Shoes|    Fashion| 3500|          2|2025-01-
18|Chennai|    7000|    Priya| 27|Chennai|
|          7|    110|    Laptop|Electronics|60000|          1|2025-01-
19|Kolkata|    60000|    Amit| 40|Kolkata|
|          8|    111|    Watch|    Fashion| 4500|          2|2025-01-
20|Delhi|    9000|    Meena| 26|Delhi|
|          4|    112|    Mobile|Electronics|28000|          1|2025-01-
21|Mumbai|    28000|    Sneha| 35|Mumbai|
|          9|    113|    TV|Electronics|55000|          1|2025-01-
22|Hyderabad|    55000|    Ravi| 31|Hyderabad|
```

23	10	114	AC	Appliances	42000	1	2025-01-
	Pune	42000	Pooja	33	Pune		
	5	115	Shoes	Fashion	2500	4	2025-01-
24	Delhi	10000	Arjun	32	Delhi		

```
orders_enriched_df.join(
    customers_df,
    "customer_id"
).select(
    "customer_name",
    "product"
).show()
```

customer_name	product
Sravan	Laptop
Rahul	Mobile
Priya	TV
Sravan	Shoes
Sneha	Watch
Arjun	Refrigerator
Rahul	AC
Kiran	Mobile
Priya	Shoes
Amit	Laptop
Meena	Watch
Sneha	Mobile
Ravi	TV
Pooja	AC
Arjun	Shoes

```
joined_df.select(
    "customer_name",
    "product"
).show()
```

customer_name	product
Sravan	Laptop
Rahul	Mobile
Priya	TV
Sravan	Shoes
Sneha	Watch

Arjun	Refrigerator
Rahul	AC
Kiran	Mobile
Priya	Shoes
Amit	Laptop
Meena	Watch
Sneha	Mobile
Ravi	TV
Pooja	AC
Arjun	Shoes

```
orders_enriched_df.join(
    customers_df,
    "customer_id"
).groupBy(
    "customer_name"
).agg(
    sum("total_amount").alias("total_spending")
).show()
```

customer_name	total_spending
Sravan	56000
Rahul	100000
Priya	52000
Sneha	33000
Arjun	45000
Kiran	75000
Meena	9000
Amit	60000
Ravi	55000
Pooja	42000

```
orders_enriched_df.join(
    customers_df,
    "customer_id"
).groupBy(
    "customer_name"
).agg(
    sum("total_amount").alias("total_spending")
).orderBy(
    col("total_spending").desc()
).limit(1).show()
```

```
+-----+-----+
|customer_name|total_spending|
+-----+-----+
|      Rahul|      100000|
+-----+-----+
```

```
orders_enriched_df.groupBy(
  "city"
).agg(
  avg("total_amount").alias("avg_spending")
).show()
```

```
+-----+-----+
|    city|avg_spending|
+-----+-----+
|Hyderabad|    37000.0|
|Bangalore|    50000.0|
|   Chennai|    26000.0|
|    Mumbai|    16500.0|
|     Delhi|    18000.0|
|     Pune|    58500.0|
|   Kolkata|    60000.0|
+-----+-----+
```

```
orders_enriched_df.join(
  customers_df,
  "customer_id"
).filter(
  col("category") == "Electronics"
).select(
  "customer_name"
).distinct().show()
```

```
+-----+
|customer_name|
+-----+
|      Sravan|
|      Rahul|
|      Priya|
|      Sneha|
|      Kiran|
|      Amit|
|      Ravi|
+-----+
```

Section 6: Sorting & Distinct (35-38)

```
orders_df.orderBy(
  col("price").desc()
).show()
```

```
+-----+-----+-----+-----+-----+-----+
+-----+-----+
|order_id|customer_id|      product|      category|price|quantity|
|order_date|      city|
+-----+-----+-----+-----+-----+-----+
+-----+-----+
|      110|          7|      Laptop|Electronics|60000|          1|2025-01-
19|  Kolkata|
|      113|          9|          TV|Electronics|55000|          1|2025-01-
22|Hyderabad|
|      101|          1|      Laptop|Electronics|50000|          1|2025-01-
10|Hyderabad|
|      103|          3|          TV|Electronics|45000|          1|2025-01-
12|  Chennai|
|      114|         10|          AC|  Appliances|42000|          1|2025-01-
23|    Pune|
|      107|          2|          AC|  Appliances|40000|          1|2025-01-
16|Bangalore|
|      106|          5|Refrigerator|  Appliances|35000|          1|2025-01-
15|    Delhi|
|      102|          2|      Mobile|Electronics|30000|          2|2025-01-
11|Bangalore|
|      112|          4|      Mobile|Electronics|28000|          1|2025-01-
21|  Mumbai|
|      108|          6|      Mobile|Electronics|25000|          3|2025-01-
17|    Pune|
|      105|          4|      Watch|    Fashion| 5000|          1|2025-01-
14|  Mumbai|
|      111|          8|      Watch|    Fashion| 4500|          2|2025-01-
20|    Delhi|
|      109|          3|      Shoes|    Fashion| 3500|          2|2025-01-
18|  Chennai|
|      104|          1|      Shoes|    Fashion| 3000|          2|2025-01-
13|Hyderabad|
|      115|          5|      Shoes|    Fashion| 2500|          4|2025-01-
24|    Delhi|
+-----+-----+-----+-----+-----+-----+
+-----+-----+
```

```
orders_df.select(
  "category"
).distinct().show()
```

```
+-----+
|  category|
```



```
+-----+
|Electronics|
|   Fashion|
| Appliances|
+-----+
```

```
orders_df.select("city").distinct().show()
```

```
+-----+
|   city|
+-----+
|Hyderabad|
|Bangalore|
|   Chennai|
|   Mumbai|
|   Delhi|
|   Pune|
|Kolkata|
+-----+
```

```
orders_df.orderBy(col("price").desc()).show(5)
```

```
+-----+-----+-----+-----+-----+-----+-----+
+-----+
|order_id|customer_id|product|   category|price|quantity|order_date|
city|
+-----+-----+-----+-----+-----+-----+-----+
+-----+
|   110|          7| Laptop|Electronics|60000|          1|2025-01-19|
Kolkata|
|   113|          9|   TV|Electronics|55000|          1|2025-01-22|
Hyderabad|
|   101|          1| Laptop|Electronics|50000|          1|2025-01-10|
Hyderabad|
|   103|          3|   TV|Electronics|45000|          1|2025-01-12|
Chennai|
|   114|         10|   AC| Appliances|42000|          1|2025-01-23|
Pune|
+-----+-----+-----+-----+-----+-----+-----+
+-----+
only showing top 5 rows
```

Section 7: Window Functions (39-45)

```
#window
revenue_window = Window.orderBy(
    col("total_amount").desc()
)
```

```
orders_enriched_df.withColumn(
    "row_num",
    row_number().over(revenue_window)
).show()
```

```
/databricks/python/lib/python3.12/site-packages/pyspark/sql/connect/
expressions.py:1160: UserWarning: WARN WindowExpression: No Partition
Defined for Window operation! Moving all data to a single partition,
this can cause serious performance degradation.
    warnings.warn(
```

```
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+
|order_id|customer_id|    product|    category|price|quantity|
order_date|    city|total_amount|row_num|
+-----+-----+-----+-----+
+-----+-----+-----+-----+
|    108|        6|    Mobile|Electronics|25000|    3|2025-01-
17|    Pune|    75000|    1|
|    102|        2|    Mobile|Electronics|30000|    2|2025-01-
11|Bangalore|    60000|    2|
|    110|        7|    Laptop|Electronics|60000|    1|2025-01-
19|    Kolkata|    60000|    3|
|    113|        9|        TV|Electronics|55000|    1|2025-01-
22|Hyderabad|    55000|    4|
|    101|        1|    Laptop|Electronics|50000|    1|2025-01-
10|Hyderabad|    50000|    5|
|    103|        3|        TV|Electronics|45000|    1|2025-01-
12|    Chennai|    45000|    6|
|    114|       10|        AC|    Appliances|42000|    1|2025-01-
23|    Pune|    42000|    7|
|    107|        2|        AC|    Appliances|40000|    1|2025-01-
16|Bangalore|    40000|    8|
|    106|        5|Refrigerator|    Appliances|35000|    1|2025-01-
15|    Delhi|    35000|    9|
|    112|        4|    Mobile|Electronics|28000|    1|2025-01-
21|    Mumbai|    28000|   10|
|    115|        5|    Shoes|    Fashion|    2500|    4|2025-01-
24|    Delhi|    10000|   11|
|    111|        8|    Watch|    Fashion|    4500|    2|2025-01-
20|    Delhi|     9000|   12|
|    109|        3|    Shoes|    Fashion|    3500|    2|2025-01-
18|    Chennai|     7000|   13|
|    104|        1|    Shoes|    Fashion|    3000|    2|2025-01-
13|Hyderabad|     6000|   14|
|    105|        4|    Watch|    Fashion|    5000|    1|2025-01-
14|    Mumbai|     5000|   15|
+-----+-----+-----+-----+
+-----+-----+-----+-----+
```

```
#window
category_window = Window.partitionBy(
    "category"
).orderBy(
    col("price").desc()
)
orders_df.withColumn(
    "rank",
    rank().over(category_window)
).show()
```

```
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+
|order_id|customer_id|    product|    category|price|quantity|
order_date|    city|rank|
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+
|    114|    10|    AC| Appliances|42000|    1|2025-01-
23|    Pune|    1|
|    107|    2|    AC| Appliances|40000|    1|2025-01-
16|Bangalore|    2|
|    106|    5|Refrigerator| Appliances|35000|    1|2025-01-
15|    Delhi|    3|
|    110|    7|    Laptop|Electronics|60000|    1|2025-01-
19|    Kolkata|    1|
|    113|    9|    TV|Electronics|55000|    1|2025-01-
22|Hyderabad|    2|
|    101|    1|    Laptop|Electronics|50000|    1|2025-01-
10|Hyderabad|    3|
|    103|    3|    TV|Electronics|45000|    1|2025-01-
12|    Chennai|    4|
|    102|    2|    Mobile|Electronics|30000|    2|2025-01-
11|Bangalore|    5|
|    112|    4|    Mobile|Electronics|28000|    1|2025-01-
21|    Mumbai|    6|
|    108|    6|    Mobile|Electronics|25000|    3|2025-01-
17|    Pune|    7|
|    105|    4|    Watch|    Fashion| 5000|    1|2025-01-
14|    Mumbai|    1|
|    111|    8|    Watch|    Fashion| 4500|    2|2025-01-
20|    Delhi|    2|
|    109|    3|    Shoes|    Fashion| 3500|    2|2025-01-
18|    Chennai|    3|
|    104|    1|    Shoes|    Fashion| 3000|    2|2025-01-
13|Hyderabad|    4|
|    115|    5|    Shoes|    Fashion| 2500|    4|2025-01-
24|    Delhi|    5|
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+
```

```
orders_df.withColumn(
  "dense_rank",
  dense_rank().over(category_window)
).show()
```

```
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+
|order_id|customer_id|    product|    category|price|quantity|
order_date|    city|dense_rank|
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+
|    114|    10|    AC| Appliances|42000|    1|2025-01-
23|    Pune|    1|
|    107|    2|    AC| Appliances|40000|    1|2025-01-
16|Bangalore|    2|
|    106|    5|Refrigerator| Appliances|35000|    1|2025-01-
15|    Delhi|    3|
|    110|    7|    Laptop|Electronics|60000|    1|2025-01-
19|    Kolkata|    1|
|    113|    9|    TV|Electronics|55000|    1|2025-01-
22|Hyderabad|    2|
|    101|    1|    Laptop|Electronics|50000|    1|2025-01-
10|Hyderabad|    3|
|    103|    3|    TV|Electronics|45000|    1|2025-01-
12|    Chennai|    4|
|    102|    2|    Mobile|Electronics|30000|    2|2025-01-
11|Bangalore|    5|
|    112|    4|    Mobile|Electronics|28000|    1|2025-01-
21|    Mumbai|    6|
|    108|    6|    Mobile|Electronics|25000|    3|2025-01-
17|    Pune|    7|
|    105|    4|    Watch|    Fashion| 5000|    1|2025-01-
14|    Mumbai|    1|
|    111|    8|    Watch|    Fashion| 4500|    2|2025-01-
20|    Delhi|    2|
|    109|    3|    Shoes|    Fashion| 3500|    2|2025-01-
18|    Chennai|    3|
|    104|    1|    Shoes|    Fashion| 3000|    2|2025-01-
13|Hyderabad|    4|
|    115|    5|    Shoes|    Fashion| 2500|    4|2025-01-
24|    Delhi|    5|
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+

```

```
#Find Highest Priced Product In Each Category
highest_df = orders_df.withColumn(
  "rank",
  rank().over(category_window)
)
```

```
highest_df.filter(
    col("rank") == 1
).show()
```

```
+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+
|order_id|customer_id|product|category|price|quantity|order_date|
city|rank|
+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+
|114|10|AC|Appliances|42000|1|2025-01-23|
Pune|1|
|110|7|Laptop|Electronics|60000|1|2025-01-19|
Kolkata|1|
|105|4|Watch|Fashion|5000|1|2025-01-14|
Mumbai|1|
+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+
```

```
highest_df.filter(
    col("rank") == 2
).show()
```

```
+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+
|order_id|customer_id|product|category|price|quantity|order_date|
city|rank|
+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+
|107|2|AC|Appliances|40000|1|2025-01-16|
Bangalore|2|
|113|9|TV|Electronics|55000|1|2025-01-22|
Hyderabad|2|
|111|8|Watch|Fashion|4500|2|2025-01-20|
Delhi|2|
+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+
```

```
#Calculate Running Revenue Per City
#window
city_window = Window.partitionBy(
    "city"
).orderBy(
    "order_date"
)
orders_enriched_df.withColumn(
    "running_revenue",
```

```
sum("total_amount").over(city_window)
).show()
```

```
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+
|order_id|customer_id|    product|    category|price|quantity|
order_date|    city|total_amount|running_revenue|
+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+
|    102|      2|    Mobile|Electronics|30000|      2|2025-01-
11|Bangalore|    60000|    60000|
|    107|      2|      AC|Appliances|40000|      1|2025-01-
16|Bangalore|    40000|    100000|
|    103|      3|      TV|Electronics|45000|      1|2025-01-
12|Chennai|    45000|    45000|
|    109|      3|    Shoes|Fashion|3500|      2|2025-01-
18|Chennai|    7000|    52000|
|    106|      5|Refrigerator|Appliances|35000|      1|2025-01-
15|Delhi|    35000|    35000|
|    111|      8|    Watch|Fashion|4500|      2|2025-01-
20|Delhi|    9000|    44000|
|    115|      5|    Shoes|Fashion|2500|      4|2025-01-
24|Delhi|    10000|    54000|
|    101|      1|Laptop|Electronics|50000|      1|2025-01-
10|Hyderabad|    50000|    50000|
|    104|      1|    Shoes|Fashion|3000|      2|2025-01-
13|Hyderabad|    6000|    56000|
|    113|      9|      TV|Electronics|55000|      1|2025-01-
22|Hyderabad|    55000|    111000|
|    110|      7|Laptop|Electronics|60000|      1|2025-01-
19|Kolkata|    60000|    60000|
|    105|      4|    Watch|Fashion|5000|      1|2025-01-
14|Mumbai|    5000|    5000|
|    112|      4|    Mobile|Electronics|28000|      1|2025-01-
21|Mumbai|    28000|    33000|
|    108|      6|    Mobile|Electronics|25000|      3|2025-01-
17|Pune|    75000|    75000|
|    114|     10|      AC|Appliances|42000|      1|2025-01-
23|Pune|    42000|    117000|
+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+

```

```
# Calculate Cumulative Quantity Sold
# Window
quantity_window = Window.orderBy(
    "order_date"
)
orders_df.withColumn(
    "cumulative_quantity",
```

```
sum("quantity").over(quantity_window)
).show()
```

/databricks/python/lib/python3.12/site-packages/pyspark/sql/connect/expressions.py:1160: UserWarning: WARN WindowExpression: No Partition Defined for Window operation! Moving all data to a single partition, this can cause serious performance degradation.

```
warnings.warn(
```

```
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+
|order_id|customer_id|    product|  category|price|quantity|
order_date|    city|cumulative_quantity|
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+
|    101|          1|    Laptop|Electronics|50000|    1|2025-01-
10|Hyderabad|          1|
|    102|          2|    Mobile|Electronics|30000|    2|2025-01-
11|Bangalore|          3|
|    103|          3|      TV|Electronics|45000|    1|2025-01-
12|  Chennai|          4|
|    104|          1|    Shoes|    Fashion| 3000|    2|2025-01-
13|Hyderabad|          6|
|    105|          4|    Watch|    Fashion| 5000|    1|2025-01-
14|  Mumbai|          7|
|    106|          5|Refrigerator| Appliances|35000|    1|2025-01-
15|   Delhi|          8|
|    107|          2|      AC| Appliances|40000|    1|2025-01-
16|Bangalore|          9|
|    108|          6|    Mobile|Electronics|25000|    3|2025-01-
17|   Pune|         12|
|    109|          3|    Shoes|    Fashion| 3500|    2|2025-01-
18|  Chennai|         14|
|    110|          7|    Laptop|Electronics|60000|    1|2025-01-
19| Kolkata|         15|
|    111|          8|    Watch|    Fashion| 4500|    2|2025-01-
20|   Delhi|         17|
|    112|          4|    Mobile|Electronics|28000|    1|2025-01-
21|  Mumbai|         18|
|    113|          9|      TV|Electronics|55000|    1|2025-01-
22|Hyderabad|         19|
|    114|         10|      AC| Appliances|42000|    1|2025-01-
23|   Pune|         20|
|    115|          5|    Shoes|    Fashion| 2500|    4|2025-01-
24|   Delhi|         24|
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+
```

Section 8: Advanced Transformations (46-55)

```

# Classify Orders
# Requirement
# High Value    > 50000
# Medium Value  20000 - 50000
# Low Value     < 20000
# Solution
orders_enriched_df.withColumn(
    "value_category",
    when(col("total_amount") > 50000, "High Value")
    .when(
        (col("total_amount") >= 20000) &
        (col("total_amount") <= 50000),
        "Medium Value"
    )
    .otherwise("Low Value")
).show()

```

```

+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
|order_id|customer_id|    product|    category|price|quantity|
order_date|    city|total_amount|value_category|
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
|    101|      1|    Laptop|Electronics|50000|      1|2025-01-
10|Hyderabad|    50000|    Medium Value|
|    102|      2|    Mobile|Electronics|30000|      2|2025-01-
11|Bangalore|    60000|      High Value|
|    103|      3|         TV|Electronics|45000|      1|2025-01-
12|  Chennai|    45000|    Medium Value|
|    104|      1|     Shoes|   Fashion| 3000|      2|2025-01-
13|Hyderabad|    6000|     Low Value|
|    105|      4|     Watch|   Fashion| 5000|      1|2025-01-
14|  Mumbai|    5000|     Low Value|
|    106|      5|Refrigerator|Appliances|35000|      1|2025-01-
15|   Delhi|    35000|    Medium Value|
|    107|      2|         AC|Appliances|40000|      1|2025-01-
16|Bangalore|    40000|    Medium Value|
|    108|      6|    Mobile|Electronics|25000|      3|2025-01-
17|    Pune|    75000|      High Value|
|    109|      3|     Shoes|   Fashion| 3500|      2|2025-01-
18|  Chennai|    7000|     Low Value|
|    110|      7|    Laptop|Electronics|60000|      1|2025-01-
19| Kolkata|    60000|      High Value|
|    111|      8|     Watch|   Fashion| 4500|      2|2025-01-
20|   Delhi|    9000|     Low Value|
|    112|      4|    Mobile|Electronics|28000|      1|2025-01-
21|  Mumbai|    28000|    Medium Value|
|    113|      9|         TV|Electronics|55000|      1|2025-01-
22|Hyderabad|    55000|      High Value|
|    114|     10|         AC|Appliances|42000|      1|2025-01-

```



```

23|      Pune|      42000| Medium Value|
|      115|      5|      Shoes|      Fashion| 2500|      4|2025-01-
24|      Delhi|      10000|      Low Value|
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+

```

Remove Duplicate Products

```

orders_df.dropDuplicates(
    ["product"]
).show()

```

```

+-----+-----+-----+-----+-----+-----+
+-----+-----+
|order_id|customer_id|      product|      category|price|quantity|
order_date|      city|
+-----+-----+-----+-----+-----+-----+
+-----+-----+
|      101|      1|      Laptop|Electronics|50000|      1|2025-01-
10|Hyderabad|
|      102|      2|      Mobile|Electronics|30000|      2|2025-01-
11|Bangalore|
|      103|      3|      TV|Electronics|45000|      1|2025-01-
12|Chennai|
|      104|      1|      Shoes|      Fashion| 3000|      2|2025-01-
13|Hyderabad|
|      105|      4|      Watch|      Fashion| 5000|      1|2025-01-
14|Mumbai|
|      106|      5|Refrigerator| Appliances|35000|      1|2025-01-
15|Delhi|
|      107|      2|      AC| Appliances|40000|      1|2025-01-
16|Bangalore|
+-----+-----+-----+-----+-----+-----+
+-----+-----+

```

Pivot Category-Wise Sales By City

Requirement

Convert:

City Category Revenue

into:

City Electronics Fashion Appliances

Solution

```

pivot_df = orders_enriched_df.groupBy("city") \
    .pivot("category") \
    .agg(sum("total_amount"))
pivot_df.show()

```

```

+-----+-----+-----+-----+
|      city|Appliances|Electronics|Fashion|

```

Hyderabad	NULL	105000	6000
Bangalore	40000	60000	NULL
Chennai	NULL	45000	7000
Mumbai	NULL	28000	5000
Delhi	35000	NULL	19000
Pune	42000	75000	NULL
Kolkata	NULL	60000	NULL

```
print(type(pivot_df))
```

```
<class 'pyspark.sql.connect.dataframe.DataFrame'>
```

```
#unpivot the pivoted data
```

```
unpivot_df = pivot_df.selectExpr(
    "city",
    "stack(3, \
    'Electronics', Electronics, \
    'Fashion', Fashion, \
    'Appliances', Appliances) \
    as (category,revenue)"
)
```

```
unpivot_df.show()
```

city	category	revenue
Hyderabad	Electronics	105000
Hyderabad	Fashion	6000
Hyderabad	Appliances	NULL
Chennai	Electronics	45000
Chennai	Fashion	7000
Chennai	Appliances	NULL
Bangalore	Electronics	60000
Bangalore	Fashion	NULL
Bangalore	Appliances	40000
Mumbai	Electronics	28000
Mumbai	Fashion	5000
Mumbai	Appliances	NULL
Delhi	Electronics	NULL
Delhi	Fashion	19000
Delhi	Appliances	35000
Pune	Electronics	75000
Pune	Fashion	NULL
Pune	Appliances	42000
Kolkata	Electronics	60000
Kolkata	Fashion	NULL

```
+-----+-----+-----+
only showing top 20 rows
```

```
#Create Array Column
```

```
array_df = orders_df.withColumn(
    "product_info",
    array(
        col("product"),
        col("category")
    )
).show(truncate=False)
```

```
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+
|order_id|customer_id|product      |category  |price|quantity|
order_date|city       |product_info          |
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+
|101      |1          |Laptop        |Electronics|50000|1        |2025-01-
10|Hyderabad|[[Laptop, Electronics]|
|102      |2          |Mobile        |Electronics|30000|2        |2025-01-
11|Bangalore|[Mobile, Electronics]|
|103      |3          |TV            |Electronics|45000|1        |2025-01-
12|Chennai  |[[TV, Electronics]|
|104      |1          |Shoes         |Fashion    |3000  |2        |2025-01-
13|Hyderabad|[Shoes, Fashion]|
|105      |4          |Watch         |Fashion    |5000  |1        |2025-01-
14|Mumbai   |[[Watch, Fashion]|
|106      |5          |Refrigerator  |Appliances |35000|1        |2025-01-
15|Delhi     |[[Refrigerator, Appliances]|
|107      |2          |AC            |Appliances |40000|1        |2025-01-
16|Bangalore|[AC, Appliances]|
|108      |6          |Mobile        |Electronics|25000|3        |2025-01-
17|Pune      |[[Mobile, Electronics]|
|109      |3          |Shoes         |Fashion    |3500  |2        |2025-01-
18|Chennai  |[[Shoes, Fashion]|
|110      |7          |Laptop        |Electronics|60000|1        |2025-01-
19|Kolkata   |[[Laptop, Electronics]|
|111      |8          |Watch         |Fashion    |4500  |2        |2025-01-
20|Delhi     |[[Watch, Fashion]|
|112      |4          |Mobile        |Electronics|28000|1        |2025-01-
21|Mumbai   |[[Mobile, Electronics]|
|113      |9          |TV            |Electronics|55000|1        |2025-01-
22|Hyderabad|[TV, Electronics]|
|114      |10         |AC            |Appliances |42000|1        |2025-01-
23|Pune      |[[AC, Appliances]|
|115      |5          |Shoes         |Fashion    |2500  |4        |2025-01-
24|Delhi     |[[Shoes, Fashion]|
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+

```

```
+-----+-----+-----+
```

```
# Explode Array Column
```

```
array_df = orders_df.withColumn(  
    "product_info",  
    array(  
        col("product"),  
        col("category")  
    )  
)  
array_df.select(  
    "order_id",  
    explode("product_info")  
) .show()
```

```
+-----+-----+  
|order_id|      col|  
+-----+-----+  
|      101|    Laptop|  
|      101| Electronics|  
|      102|    Mobile|  
|      102| Electronics|  
|      103|        TV|  
|      103| Electronics|  
|      104|     Shoes|  
|      104|    Fashion|  
|      105|     Watch|  
|      105|    Fashion|  
|      106|Refrigerator|  
|      106|  Appliances|  
|      107|         AC|  
|      107|  Appliances|  
|      108|    Mobile|  
|      108| Electronics|  
|      109|     Shoes|  
|      109|    Fashion|  
|      110|    Laptop|  
|      110| Electronics|  
+-----+-----+
```

only showing top 20 rows

```
#create map column
```

```
orders_df.withColumn(  
    "product_map",  
    create_map(  
        lit("Product"),  
        col("product"),  
        lit("Category"),
```

```

        col("category")
    )
).show(truncate=False)
+-----+-----+-----+-----+-----+-----+
+-----+-----+
+-----+
|order_id|customer_id|product      |category   |price|quantity|
order_date|city       |product_map
|
+-----+-----+-----+-----+-----+-----+
+-----+-----+
+-----+
|101      |1          |Laptop       |Electronics|50000|1        |2025-01-
10|Hyderabad|{Product -> Laptop, Category -> Electronics}|
|102      |2          |Mobile       |Electronics|30000|2        |2025-01-
11|Bangalore|{Product -> Mobile, Category -> Electronics}|
|103      |3          |TV           |Electronics|45000|1        |2025-01-
12|Chennai  |{Product -> TV, Category -> Electronics}|
|104      |1          |Shoes        |Fashion    |3000  |2        |2025-01-
13|Hyderabad|{Product -> Shoes, Category -> Fashion}|
|105      |4          |Watch        |Fashion    |5000  |1        |2025-01-
14|Mumbai   |{Product -> Watch, Category -> Fashion}|
|106      |5          |Refrigerator|Appliances |35000|1        |2025-01-
15|Delhi    |{Product -> Refrigerator, Category -> Appliances}|
|107      |2          |AC           |Appliances |40000|1        |2025-01-
16|Bangalore|{Product -> AC, Category -> Appliances}|
|108      |6          |Mobile       |Electronics|25000|3        |2025-01-
17|Pune     |{Product -> Mobile, Category -> Electronics}|
|109      |3          |Shoes        |Fashion    |3500  |2        |2025-01-
18|Chennai  |{Product -> Shoes, Category -> Fashion}|
|110      |7          |Laptop       |Electronics|60000|1        |2025-01-
19|Kolkata  |{Product -> Laptop, Category -> Electronics}|
|111      |8          |Watch        |Fashion    |4500  |2        |2025-01-
20|Delhi    |{Product -> Watch, Category -> Fashion}|
|112      |4          |Mobile       |Electronics|28000|1        |2025-01-
21|Mumbai   |{Product -> Mobile, Category -> Electronics}|
|113      |9          |TV           |Electronics|55000|1        |2025-01-
22|Hyderabad|{Product -> TV, Category -> Electronics}|
|114      |10         |AC           |Appliances |42000|1        |2025-01-
23|Pune     |{Product -> AC, Category -> Appliances}|
|115      |5          |Shoes        |Fashion    |2500  |4        |2025-01-
24|Delhi    |{Product -> Shoes, Category -> Fashion}|
+-----+-----+-----+-----+-----+-----+
+-----+-----+
+-----+

```

Section 9: Real-Time Data Engineer Scenarios (56-65)

#Find Customers Purchasing From Multiple Categories

```
orders_df.join(
    customers_df,
    "customer_id"
).groupBy(
    "customer_name"
).agg(
    countDistinct("category").alias("category_count")
).filter(
    col("category_count") > 1
).show()
```

customer_name	category_count
Rahul	2
Sneha	2
Sravan	2
Arjun	2
Priya	2

Identify Customers Who Bought Both Mobile And Laptop
Solution

```
orders_df.filter(
    col("product").isin("Mobile", "Laptop")
).groupBy(
    "customer_id"
).agg(
    countDistinct("product").alias("product_count")
).filter(
    col("product_count") == 2
).show()
```

customer_id	product_count
-------------	---------------

Find Repeat Customers

```
orders_df.join(
    customers_df,
    "customer_id"
).groupBy(
    "customer_name"
).agg(
    count("order_id").alias("order_count")
).filter(
```

```
col("order_count") > 1
).show()
```

```
+-----+-----+
|customer_name|order_count|
+-----+-----+
|      Sravan|          2|
|      Rahul|          2|
|      Priya|          2|
|      Sneha|          2|
|      Arjun|          2|
+-----+-----+
```

Category Contribution Percentage To Total Revenue

```
category_revenue = orders_enriched_df.groupBy(
    "category"
).agg(
    sum("total_amount").alias("revenue")
)
total_revenue = orders_enriched_df.agg(
    sum("total_amount")
).collect()[0][0]
category_revenue.withColumn(
    "contribution_pct",
    round(
        col("revenue") * 100 / total_revenue,
        2
    )
).show()
```

```
+-----+-----+-----+
|category|revenue|contribution_pct|
+-----+-----+-----+
|Electronics| 373000|          70.78|
|    Fashion|  37000|           7.02|
|Appliances| 117000|          22.2|
+-----+-----+-----+
```

Monthly Sales Trend

Convert Date

```
orders_enriched_df = orders_enriched_df.withColumn(
    "order_date",
    to_date("order_date")
)
```

Solution

```
orders_enriched_df.groupBy(
    month("order_date").alias("month")
).agg(
```

```

    sum("total_amount").alias("monthly_sales")
).orderBy(
    "month"
).show()

```

```

+-----+-----+
|month|monthly_sales|
+-----+-----+
|    1|         527000|
+-----+-----+

```

```

orders_df.groupBy(
    "product"
).agg(
    sum("quantity").alias("total_qty")
).orderBy(
    col("total_qty").desc()
).show(1)

```

```

+-----+-----+
|product|total_qty|
+-----+-----+
|  Shoes|         8|
+-----+-----+

```

only showing top 1 row

```

orders_df.groupBy(
    "product"
).agg(
    sum("quantity").alias("total_qty")
).orderBy(
    col("total_qty").asc()
).show(1)

```

```

+-----+-----+
|    product|total_qty|
+-----+-----+
|Refrigerator|         1|
+-----+-----+

```

only showing top 1 row

```

# Customer Retention Percentage
# Simple Definition
# Repeat Customers / Total Customers
# Step 1
# Repeat Customers
repeat_customers = orders_df.groupBy(
    "customer_id"
).count().filter(
    col("count") > 1
)

```



```

).count()
# Step 2
# Total Customers
total_customers = orders_df.select(
    "customer_id"
).distinct().count()
# Step 3
# Retention %
retention_pct = (
    repeat_customers * 100
) / total_customers

print(retention_pct)

50.0

# City-Wise Revenue Contribution
# Revenue By City
city_revenue = orders_enriched_df.groupBy(
    "city"
).agg(
    sum("total_amount").alias("revenue")
)
# Total Revenue
total_revenue = orders_enriched_df.agg(
    sum("total_amount")
).collect()[0][0]
# Contribution %
city_revenue.withColumn(
    "contribution_pct",
    round(
        col("revenue") * 100 / total_revenue,
        2
    )
).show()

```

city	revenue	contribution_pct
Hyderabad	111000	21.06
Bangalore	100000	18.98
Chennai	52000	9.87
Mumbai	33000	6.26
Delhi	54000	10.25
Pune	117000	22.2
Kolkata	60000	11.39

```

# Generate Customer Segmentation Report
# Business Rule
# Premium > 100000
# Gold    50000 - 100000
# Silver  < 50000
# Step 1
# Revenue Per Customer

customer_revenue = orders_enriched_df.groupBy(
    "customer_id"
).agg(
    sum("total_amount").alias("revenue")
)
# Step 2
# Segment
customer_revenue.withColumn(
    "segment",
    when(
        col("revenue") > 100000,
        "Premium"
    )
    .when(
        col("revenue") >= 50000,
        "Gold"
    )
    .otherwise("Silver")
).show()

```

```

+-----+-----+-----+
|customer_id|revenue|segment|
+-----+-----+-----+
|          1|  56000|   Gold|
|          2| 100000|   Gold|
|          3|  52000|   Gold|
|          4|  33000|  Silver|
|          5|  45000|  Silver|
|          6|  75000|   Gold|
|          7|  60000|   Gold|
|          8|   9000|  Silver|
|          9|  55000|   Gold|
|         10|  42000|  Silver|
+-----+-----+-----+

```

Bonus Interview Questions (66-75)

1. Difference between select() and withColumn()?

Answer: select() is used to choose specific columns from a DataFrame, whereas withColumn() is used to create a new column or modify an existing column.

1. Difference between `distinct()` and `dropDuplicates()`?

Answer: `distinct()` removes duplicate rows based on all columns, whereas `dropDuplicates()` removes duplicates based on specific columns provided.

1. Difference between `groupBy()` and Window Functions?

Answer: `groupBy()` aggregates data and reduces the number of rows, whereas Window Functions perform calculations while preserving all original rows.

1. When would you use Broadcast Join?

Answer: Broadcast Join is used when one table is small enough to fit into memory and the other table is very large. It avoids shuffling and improves join performance.

1. Difference between `repartition()` and `coalesce()`?

Answer: `repartition()` can increase or decrease partitions and causes a full shuffle. `coalesce()` is mainly used to reduce partitions and avoids a full shuffle, making it more efficient.

1. What causes Data Skew?

Answer: Data skew occurs when some partition keys contain significantly more data than others, causing uneven workload distribution across executors.

1. How would you optimize a large join?

Answer: Optimize large joins by broadcasting small tables, filtering data before joins, selecting only required columns, repartitioning on join keys, and handling skewed data.

1. How do you handle Small Files Problem?

Answer: Handle small files by using `repartition/coalesce` before writing data and using Delta Lake `OPTIMIZE` to compact files into larger files.

1. Difference between Narrow and Wide Transformations?

Answer: Narrow transformations do not require data shuffling between partitions, whereas Wide transformations require shuffling data across partitions and are more expensive.

1. How does Catalyst Optimizer improve performance?

Answer: Catalyst Optimizer improves performance by automatically optimizing Spark queries through predicate pushdown, column pruning, join optimization, and efficient execution planning.